An implementation of distance marginalization

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This document describes an implementation of distance marginalization. See the Appendix C.3 of [1] for the basics of distance marginalization.

1 Basics

The logarithm of likelihood ratio has the following form,

$$\ln \mathcal{L} = a\frac{D_{\rm ref}}{D_{\rm L}} - \frac{b}{2}\left(\frac{D_{\rm ref}}{D_{\rm L}}\right)^2. \tag{1}$$

Here a and b correspond to $\kappa^2(D_{\rm ref})$ and $\rho^2_{\rm opt}(D_{\rm ref})$ in [1]. We want to calculate likelihood ratio marginalized over distance,

$$\mathcal{L}^{\rm marg} = \frac{3}{D^3_{\rm max} - D^3_{\rm min}}\int_{D_{\rm min}}^{D_{\rm max}} D_{\rm L}^2 dD_{\rm L}\mathcal{L}. \tag{2}$$

Using the substitution $x = D_{\rm ref}/D_{\rm L}$, we obtain

$$\mathcal{L}^{\rm marg} = \frac{3D^3_{\rm ref}}{D^3_{\rm max} - D^3_{\rm min}}\int_{x_{\rm min}}^{x_{\rm max}} x^{-4}{\rm e}^{f(x)}dx, \tag{3}$$

where

$$f(x) \equiv -\frac{b}{2}x^2 + ax, \quad x_{\rm min} \equiv \frac{D_{\rm ref}}{D_{\rm max}}, \quad x_{\rm max} \equiv \frac{D_{\rm ref}}{D_{\rm min}}. \tag{4}$$

Completing the square for the exponent, we obtain

$$f(x) = -\frac{b}{2}(x - x_0)^2 + \frac{b}{2}x_0^2, \tag{5}$$

where $x_0 \equiv a/b$. Thus, its maximum in $x_{\rm min} \leq x \leq x_{\rm max}$ is given by

$$f_{\rm max}(x_0, b) = \begin{cases} \dfrac{b}{2}x_0^2 - \dfrac{b}{2}(x_{\rm min} - x_0)^2, & (x_0 < x_{\rm min}) \\ \dfrac{b}{2}x_0^2, & (x_{\rm min} \leq x_0 \leq x_{\rm max}) \\ \dfrac{b}{2}x_0^2 - \dfrac{b}{2}(x_0 - x_{\rm max})^2. & (x_0 > x_{\rm max}) \end{cases} \tag{6}$$

To avoid the numerical overflow in the numerical integration, we subtract the maximum from the exponent,

$$\mathcal{L}^{\rm marg} = \frac{3D_{\rm ref}^3}{D_{\rm max}^3 - D_{\rm min}^3} {\rm e}^{f_{\rm max}(x_0,b)} I(x_0,b), \qquad I(x_0,b) \equiv \int_{x_{\rm min}}^{x_{\rm max}} x^{-4} {\rm e}^{f(x)-f_{\rm max}(x_0,b)} dx, \tag{7}$$

and numerically compute $I(x_0,b)$. The logarithm of $\mathcal{L}^{\rm marg}$ is obtained by

$$\ln \mathcal{L}^{\rm marg} = f_{\rm max}(x_0,b) + \ln\left(\frac{3D_{\rm ref}^3}{D_{\rm max}^3 - D_{\rm min}^3}\right) + \ln I(x_0,b). \tag{8}$$

2 Numerical integration with respect to distance

Next, we describe an efficient way of numerical integration with respect to distance. For large b, the integrand has a sharp peak around the maximum (See Fig. 1), which requires extremely fine grid points to resolve it. On the other hand, $I(x_0,b)$ is almost vanishing except for around the maximum. Thus, we can approximate the integral as an integral over a very narrow interval around the maximum. To be more concrete, we make the approximation $I(x_0,b) \simeq I_{\delta x}(x_0,b)$, where

$$I_{\delta x}(x_0,b) \equiv \begin{cases} \displaystyle\int_{x_{\rm min}}^{\min[x_{\rm min}+\delta x, x_{\rm max}]} x^{-4} {\rm e}^{f(x)-f_{\rm max}(x_0,b)} dx, & (x_0 < x_{\rm min}) \\ \displaystyle\int_{\max[x_0-\delta x, x_{\rm min}]}^{\min[x_0+\delta x, x_{\rm max}]} x^{-4} {\rm e}^{f(x)-f_{\rm max}(x_0,b)} dx, & (x_{\rm min} \le x_0 \le x_{\rm max}) \\ \displaystyle\int_{\max[x_{\rm max}-\delta x, x_{\rm min}]}^{x_{\rm max}} x^{-4} {\rm e}^{f(x)-f_{\rm max}(x_0,b)} dx. & (x_0 > x_{\rm max}) \end{cases} \tag{9}$$

The width of the integration interval δx is determined so that the relative error is smaller than a used-specified tolerance ϵ,

$$\frac{|I(x_0,b) - I_{\delta x}(x_0,b)|}{I(x_0,b)} \le \epsilon. \tag{10}$$

We explain how to obtain such δx in the following.

$\underline{x_0 < x_{\min}}$

For the numerator of (10), we have the following inequality,

$$\begin{aligned}
|I - I_{\delta x}| &= \int_{\min[x_{\min}+\delta x, x_{\max}]}^{x_{\max}} x^{-4} \mathrm{e}^{f(x)-f_{\max}} dx \\
&\le x_{\min}^{-4} \int_{x_{\min}+\delta x}^{\infty} \mathrm{e}^{f(x)-f_{\max}} dx \\
&= x_{\min}^{-4} \sqrt{\frac{\pi}{2b}} \mathrm{e}^{\frac{b}{2}(x_{\min}-x_0)^2} \mathrm{erfc}\left(\sqrt{\frac{b}{2}}(x_{\min}+\delta x - x_0)\right) \\
&= x_{\min}^{-4} \sqrt{\frac{\pi}{2b}} \mathrm{e}^{-\frac{b}{2}\delta x(2x_{\min}-2x_0+\delta x)} \mathrm{erfcx}\left(\sqrt{\frac{b}{2}}\,(x_{\min}-x_0+\delta x)\right) \\
&\le x_{\min}^{-4} \sqrt{\frac{\pi}{2b}} \mathrm{e}^{-\frac{b}{2}\delta x(2x_{\min}-2x_0+\delta x)}, \qquad (11)
\end{aligned}$$

where

$$\mathrm{erfc}(z) \equiv \frac{2}{\sqrt{\pi}} \int_z^{\infty} \mathrm{e}^{-t^2} dt, \qquad \mathrm{erfcx}(z) \equiv \mathrm{e}^{z^2} \mathrm{erfc}(z). \qquad (12)$$

From the fourth line to the fifth line, we have used the fact that $\mathrm{erfcx}(z)$ is simply decreasing and

$$\mathrm{erfcx}(z) < \mathrm{erfcx}(0) = 1 \qquad (13)$$

for positive z.

The denominator of (10) is given by

$$I = \int_{x_{\min}}^{x_{\max}} \mathrm{e}^{-\frac{b}{2}(x-x_0)^2 + \frac{b}{2}(x_{\min}-x_0)^2 - 4\ln x} dx. \qquad (14)$$

Using the following inequality,

$$\ln x \le \ln x_{\min} + \frac{x - x_{\min}}{x_{\min}}, \qquad (15)$$

which follows from the convexity of $y = \ln x$, we obtain the following inequality,

$$\begin{aligned}
I &\ge \int_{x_{\min}}^{x_{\max}} \mathrm{e}^{-\frac{b}{2}(x-x_0)^2 + \frac{b}{2}(x_{\min}-x_0)^2 - 4\ln x_{\min} - \frac{4}{x_{\min}}(x-x_{\min})} dx \\
&= x_{\min}^{-4} \sqrt{\frac{\pi}{2b}} \left(\mathrm{erfcx}(\alpha_1) - \mathrm{e}^{-(\alpha_2^2-\alpha_1^2)} \mathrm{erfcx}(\alpha_2)\right), \qquad (16)
\end{aligned}$$

where

$$\alpha_1 \equiv \sqrt{\frac{b}{2}} \left(x_{\min} - x_0 + \frac{4}{b x_{\min}}\right), \quad \alpha_2 \equiv \sqrt{\frac{b}{2}} \left(x_{\max} - x_0 + \frac{4}{b x_{\min}}\right). \qquad (17)$$

Thus, we have

$$\frac{|I - I_{\delta x}|}{I} \leq \frac{\mathrm{e}^{-\frac{b}{2}\delta x(2x_{\min}-2x_0+\delta x)}}{\mathrm{erfcx}(\alpha_1) - \mathrm{e}^{-(\alpha_2^2-\alpha_1^2)}\mathrm{erfcx}(\alpha_2)}, \tag{18}$$

and the following value of δx satisfies (10),

$$\delta x = \frac{2A}{\sqrt{b^2(x_{\min}-x_0)^2 + 2bA} + b(x_{\min}-x_0)}, \tag{19}$$

$$A \equiv -\ln\left[\epsilon\left(\mathrm{erfcx}(\alpha_1) - \mathrm{e}^{-(\alpha_2^2-\alpha_1^2)}\mathrm{erfcx}(\alpha_2)\right)\right]. \tag{20}$$

Since $A = \mathcal{O}(1)$, we have

$$\delta x \sim \min\left[\frac{1}{b(x_{\min}-x_0)}, \frac{1}{\sqrt{b}}\right]. \tag{21}$$

It can be easily shown that this is in the same order with the width of the peak, allowing us to use a fixed number of grid points to accurately calculate $I(x_0, b)$ even for large b.

$\underline{x_{\min} \leq x_0 \leq x_{\max}}$

Using similar tricks to what have been done for $x_0 < x_{\min}$, we obtain

$$\begin{aligned}
|I - I_{\delta x}| &= \int_{x_{\min}}^{\max[x_0-\delta x, x_{\min}]} x^{-4}\mathrm{e}^{-\frac{b}{2}(x-x_0)^2}dx + \int_{\min[x_0+\delta x, x_{\max}]}^{x_{\max}} x^{-4}\mathrm{e}^{-\frac{b}{2}(x-x_0)^2}dx \\
&\leq x_{\min}^{-4}\left[\int_{-\infty}^{x_0-\delta x} \mathrm{e}^{-\frac{b}{2}(x-x_0)^2}dx + \int_{x_0+\delta x}^{\infty} \mathrm{e}^{-\frac{b}{2}(x-x_0)^2}dx\right] \\
&= x_{\min}^{-4}\sqrt{\frac{2\pi}{b}}\mathrm{erfc}\left(\sqrt{\frac{b}{2}}\delta x\right),
\end{aligned} \tag{22}$$

for the numerator of (10), and

$$\begin{aligned}
I &\geq \int_{x_{\min}}^{x_{\max}} \mathrm{e}^{-\frac{b}{2}(x-x_0)^2 - 4\ln x_0 - \frac{4}{x_0}(x-x_0)}dx \\
&= x_0^{-4}\sqrt{\frac{\pi}{2b}}\mathrm{e}^{\frac{8}{bx_0^2}}\left[\mathrm{erf}\left(\sqrt{\frac{b}{2}}\left(x_{\max}-x_0+\frac{4}{bx_0}\right)\right) - \mathrm{erf}\left(\sqrt{\frac{b}{2}}\left(x_{\min}-x_0+\frac{4}{bx_0}\right)\right)\right], \\
&\geq x_0^{-4}\sqrt{\frac{\pi}{2b}}\left(\mathrm{erf}(\beta_2) - \mathrm{erf}(\beta_1)\right),
\end{aligned} \tag{23}$$

for the denominator, where

$$\beta_1 \equiv \sqrt{\frac{b}{2}}\left(x_{\min}-x_0+\frac{4}{bx_0}\right), \quad \beta_2 \equiv \sqrt{\frac{b}{2}}\left(x_{\max}-x_0+\frac{4}{bx_0}\right), \tag{24}$$

and

$$\mathrm{erf}(x) \equiv \frac{2}{\sqrt{\pi}} \int_0^x \mathrm{e}^{-t^2} dt. \tag{25}$$

Thus, the following value of δx satisfies (10),

$$\delta x = \sqrt{\frac{2}{b}} \mathrm{erfcinv}\left[\frac{\epsilon}{2} \left(\frac{x_{\min}}{x_0} \right)^4 (\mathrm{erf}(\beta_2) - \mathrm{erf}(\beta_1)) \right], \tag{26}$$

where $\mathrm{erfcinv}(z)$ is the inverse of $\mathrm{erfc}(z)$.

$\underline{x_0 > x_{\max}}$

Using similar tricks to what have been done for $x_0 < x_{\min}$, we obtain

$$\begin{aligned}
|I - I_{\delta x}| &= \int_{x_{\min}}^{\max[x_{\max}-\delta x, x_{\min}]} x^{-4} \mathrm{e}^{f(x)-f_{\max}} dx \\
&\leq x_{\min}^{-4} \int_{-\infty}^{x_{\max}-\delta x} \mathrm{e}^{f(x)-f_{\max}} dx \\
&= x_{\min}^{-4} \sqrt{\frac{\pi}{2b}} \mathrm{e}^{\frac{b}{2}(x_0-x_{\max})^2} \mathrm{erfc}\left(\sqrt{\frac{b}{2}} (x_0 - x_{\max} + \delta x) \right) \\
&= x_{\min}^{-4} \sqrt{\frac{\pi}{2b}} \mathrm{e}^{-\frac{b}{2}\delta x(2x_0-2x_{\max}+\delta x)} \mathrm{erfcx}\left(\sqrt{\frac{b}{2}} (x_0 - x_{\max} + \delta x) \right) \\
&\leq x_{\min}^{-4} \sqrt{\frac{\pi}{2b}} \mathrm{e}^{-\frac{b}{2}\delta x(2x_0-2x_{\max}+\delta x)},
\end{aligned} \tag{27}$$

for the numerator, and

$$I \geq \int_{x_{\min}}^{x_{\max}} \mathrm{e}^{-\frac{b}{2}(x_0-x)^2 + \frac{b}{2}(x_0-x_{\max})^2 - 4\ln x_{\max} - \frac{4}{x_{\max}}(x-x_{\max})} dx \tag{28}$$

$$= x_{\max}^{-4} \sqrt{\frac{\pi}{2b}} \left(\mathrm{erfcx}(\gamma_1) - \mathrm{e}^{-(\gamma_2^2-\gamma_1^2)} \mathrm{erfcx}(\gamma_2) \right), \tag{29}$$

for the denominator, where

$$\gamma_1 = \sqrt{\frac{b}{2}} \left(x_0 - x_{\max} - \frac{4}{b x_{\max}} \right), \quad \gamma_2 = \sqrt{\frac{b}{2}} \left(x_0 - x_{\min} - \frac{4}{b x_{\max}} \right). \tag{30}$$

Thus, the following value of δx satisfies (10),

$$\delta x = \frac{2C}{\sqrt{b^2(x_0 - x_{\max})^2 + 2bC} + b(x_0 - x_{\max})}, \tag{31}$$

$$C \equiv -\ln\left[\epsilon \frac{x_{\min}^4}{x_{\max}^4} \left(\mathrm{erfcx}(\gamma_1) - \mathrm{e}^{-(\gamma_2^2-\gamma_1^2)} \mathrm{erfcx}(\gamma_2) \right) \right]. \tag{32}$$

3 Interpolation of a look-up table

Next, we describe an efficient way of interpolating the look-up table of $\ln I(x_0, b)$. Let's assume that we want to calculate it in finite ranges of x_0 and b,

$$x_{0,\min} < x_0 < x_{0,\max}, \quad b < b_{\max}, \tag{33}$$

where $x_{0,\min} \leq x_{\min}$ and $x_{0,\max} \geq x_{\max}$. For efficiently interpolating the look-up table, we want to use regular grid points. This significantly reduces the computational cost of searching for interpolation nodes. Since regular grid points in the x_0–b coordinate system are not efficient to capture the behavior of $\ln I(x_0, b)$, we consider a different coordinate system.

First, we study the asymptotic behavior of $\ln I(x_0, b)$. For $x_{\min} \leq x_0 \leq x_{\max}$ and $\sqrt{b}x_0 \gg 1$, it can be approximated as

$$\ln I \simeq \ln \left[x_0^{-4} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}|f''(x_0)|(x-x_0)^2} \right] = -4 \ln x_0 - \frac{1}{2} \ln b + \frac{1}{2} \ln(2\pi). \tag{34}$$

In addition, it can be approximated as

$$\ln I \simeq \ln \left[x_{\min}^{-4} \int_{x_{\min}}^{\infty} \mathrm{e}^{-|f'(x_{\min})|(x-x_{\min})} dx \right] = -\ln(x_{\min} - x_0) - \ln b - 4 \ln x_{\min}, \tag{35}$$

for $b(x_{\min} - x_0) \gg 1$, and

$$\ln I \simeq \ln \left[x_{\max}^{-4} \int_{-\infty}^{x_{\max}} \mathrm{e}^{-|f'(x_{\max})|(x_{\max}-x)} dx \right] = -\ln(x_0 - x_{\max}) - \ln b - 4 \ln x_{\max}, \tag{36}$$

for $b(x_0 - x_{\max}) \gg 1$. Those asymptotic behaviors imply that an alternative coordinate to x_0, $s(x_0)$, should behave as

$$s(x_0) \sim \begin{cases} -\ln(x_{\min} - x_0), & (x_0 < x_{\min}) \\ \ln x_0, & (x_{\min} < x_0 < x_{\max}) \\ \ln(x_0 - x_{\max}). & (x_0 > x_{\max}) \end{cases} \tag{37}$$

On the other hand, $\ln I$ has sharp transitions at $|x_0 - x_{\min}| \lesssim 1/\sqrt{b}$ and $|x_0 - x_{\max}| \lesssim 1/\sqrt{b}$ (See the top panel of Fig. 2), which we want to make smooth in the new coordinate system.

To fulfill those requirements, we consider the following coordinate,

$$s(x_0) = \sinh^{-1}\left[\sqrt{b_{\max}}(x_0 - x_{\min})\right] - \sinh^{-1}\left[\sqrt{b_{\max}}(x_{\max} - x_0)\right]. \tag{38}$$

The bottom panel of Fig. 2 shows $\ln I$ is a smooth function with respect to s.

The asymptotic behavior of $\ln I$ implies an alternative coordinate to b should behave as $\ln b$ for large b. To cover $b = 0$, we consider the following coordinate,

$$t(b) = \sinh^{-1}\left(\frac{b}{b_{\mathrm{ref}}}\right). \tag{39}$$

Here, $b_{\rm ref}$ should be sufficiently small to satisfy $\ln I(x_0, b_{\rm ref}) \simeq \ln I(x_0, 0)$. For this to be the case, $b_{\rm ref}$ should satisfy

$$\max_{x_{\rm min} \le x_0 \le x_{\rm max}} f(x_0, b_{\rm ref}) - \min_{x_{\rm min} \le x_0 \le x_{\rm max}} f(x_0, b_{\rm ref}) \ll 1. \tag{40}$$

Thus, $b_{\rm ref}$ should be less than the inverse of the maximum of the following function,

$$d(x_0) \equiv \frac{1}{b} \left[\max_{x_{\rm min} \le x_0 \le x_{\rm max}} f(x_0, b) - \min_{x_{\rm min} \le x_0 \le x_{\rm max}} f(x_0, b) \right]. \tag{41}$$

It can be easily shown that it takes its maximum at $x_0 = x_{0,\rm min}$ or $x_0 = x_{0,\rm max}$, and we use the following value for $b_{\rm ref}$,

$$\begin{aligned} b_{\rm ref} &= \frac{1}{\max\left[d(x_{0,\rm min}), d(x_{0,\rm max})\right]} \\ &= \frac{1}{(x_{\rm max} - x_{\rm min}) \max\left[x_{\rm min} + x_{\rm max} - 2x_{0,\rm min}, 2x_{0,\rm max} - x_{\rm min} - x_{\rm max}\right]}. \end{aligned} \tag{42}$$

The look-up table is constructed over regular grid points in the s–t coordinate system with their spacings of $\Delta s \ll 1$ and $\Delta t \ll 1$.

References

[1] E. Thrane and C. Talbot, Publ. Astron. Soc. Austral. **36**, e010 (2019) [erratum: Publ. Astron. Soc. Austral. **37**, e036 (2020)] doi:10.1017/pasa.2019.2 [arXiv:1809.02293 [astro-ph.IM]].

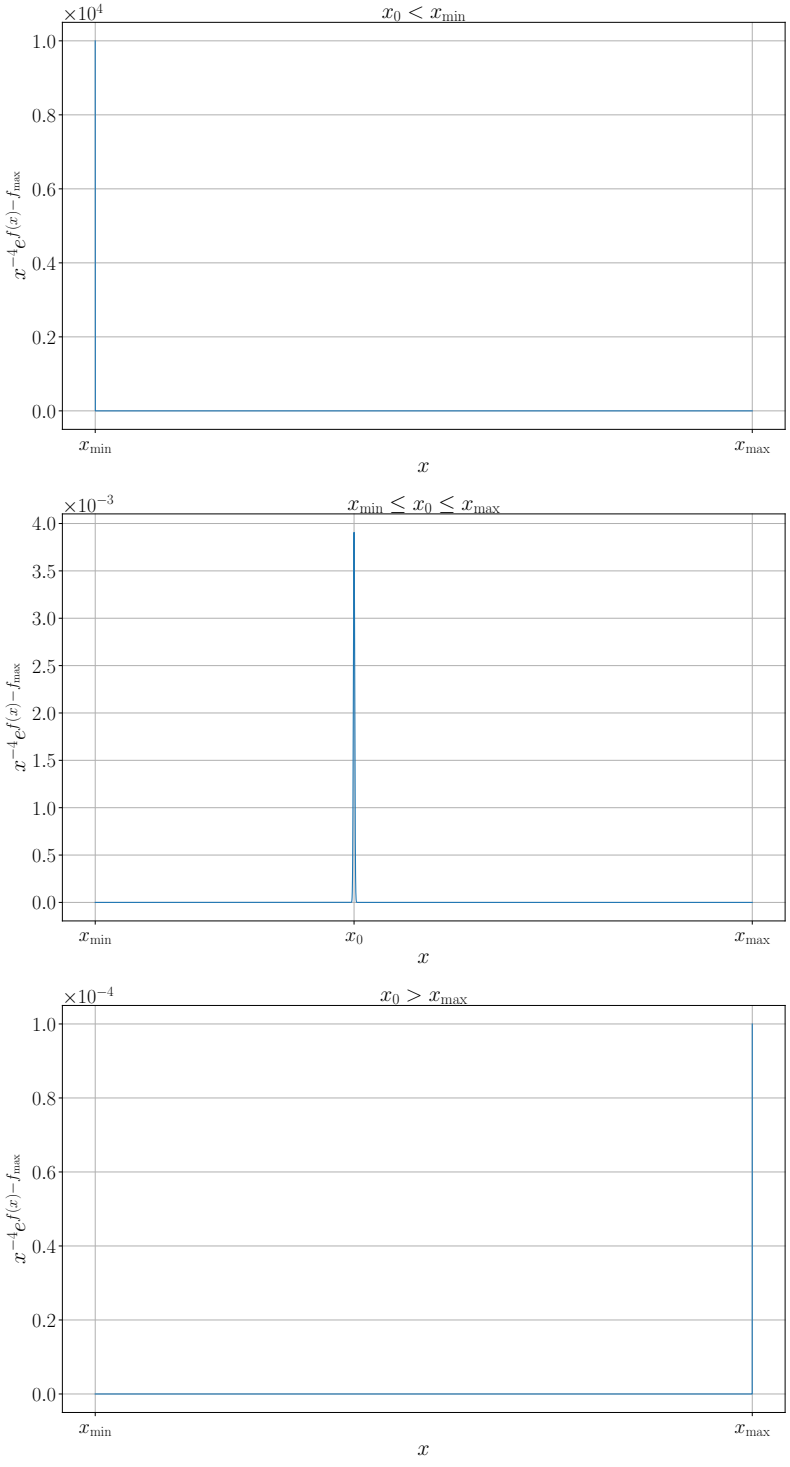


Figure 1: The integrand for $(b, x_{\min}, x_{\max}) = (10^4, 0.1, 10)$. $x_0 = -1$ (top), $x_0 = 4$ (middle), and $x_0 = 20$ (bottom).

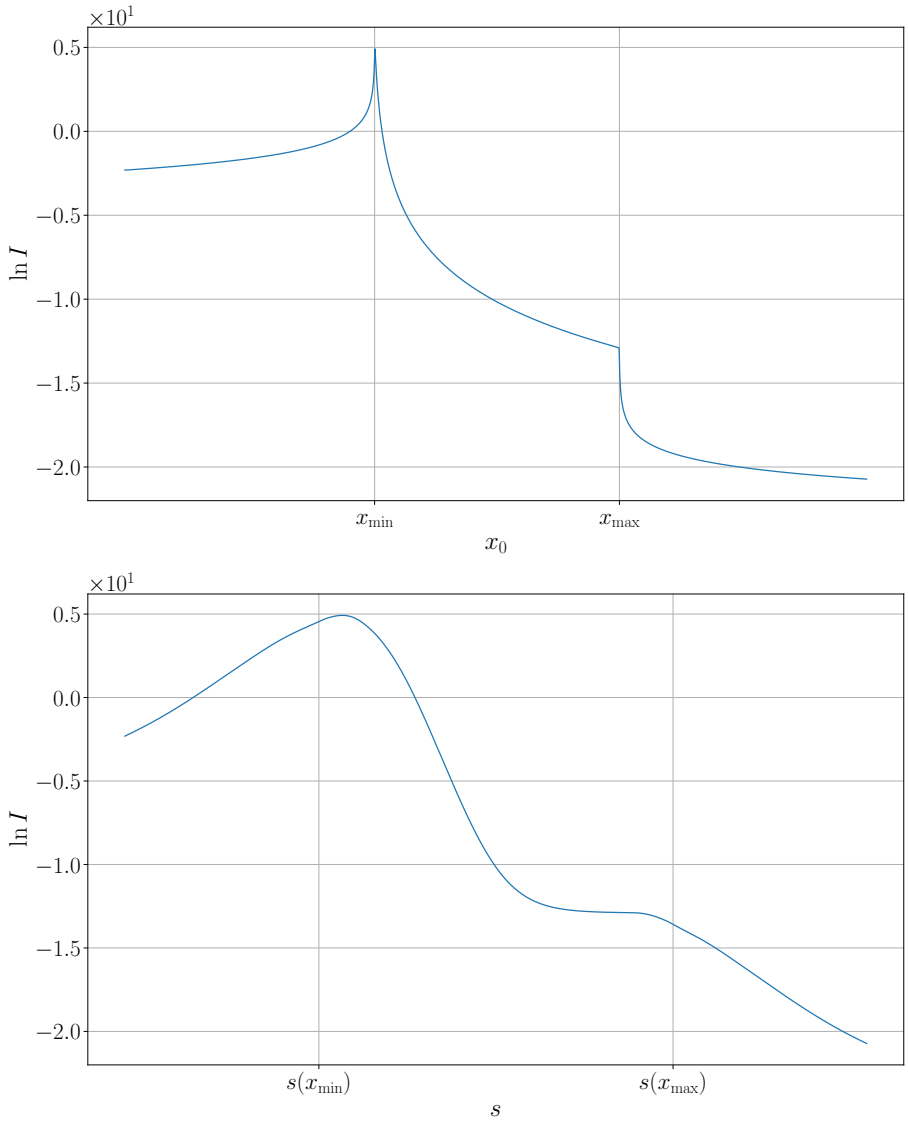


Figure 2: $\ln I(x_0, b)$ for $(b, x_{\min}, x_{\max}) = (10^4, 0.1, 10)$. The top panel shows it with respect to x_0, and bottom with respect to s with $b_{\max} = b$.